

ADITYA NAWAL

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EDUCATION

National University of Singapore (NUS)

PhD, Department of Electrical and Computer Engineering

Jan 2025 – Jan 2029

CGPA: 4.5/5.0

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering, Electronics and Electrical Engineering

Aug 2018 – June 2022

CGPA: 8.39/10

RESEARCH PUBLICATIONS

Relevance as a Vulnerability: How Web Retrieval Degrades Safety Alignment in LLM Agents [PDF]

Aditya Nawal, Manit Baser, Mohan Gurusamy

Findings of the Association for Computational Linguistics: EMNLP 2026

Oct 2026

Built AgentREVEAL to diagnose retrieval-induced safety degradation in LLM agents; identified relevance-triggered failure modes including the Safe Source Paradox, and introduced HarmURLBench with 1,405 real-world URLs and 320 harmful behaviors.

Revolutionizing visuals: The role of generative AI in modern image generation [PDF]

Gaurang Bansal, Aditya Nawal, Vinay Chamola, Norbert Herencsar

ACM Transactions on Multimedia Computing, Communications and Applications

Nov 2024

Surveyed modern generative image-generation systems and analyzed their role in multimedia content creation, including quality, diversity, interpretability, computational efficiency, and practical deployment considerations.

An IoMT-based incremental learning framework with a novel feature selection algorithm for intelligent diagnosis in smart healthcare [PDF]

Siva Sai, Kartikey Singh Bhandari, Aditya Nawal, Vinay Chamola, Biplab Sikdar

IEEE Transactions on Machine Learning in Communications and Networking

Mar 2024

Developed an IoMT-based incremental learning framework for smart healthcare diagnosis, improving anomaly detection and classification accuracy while reducing runtime versus batch-learning baselines.

ONGOING WORKS

Fetch or Refuse: What Governs Retrieval Admission

Aditya Nawal, Manit Baser, Mohan Gurusamy

Under review

TACL

Analysis of what governs whether tool-using language models admit external sources during retrieval.

SLIP-ing into Harm: Multilingual Safety Across Tool Admission and Generation in LLMs

Aditya Nawal, Manit Baser, Mohan Gurusamy

Under review

ACL ARR, Aug 2026

Cross-lingual safety behaviour of tool-using language models across the admission and generation stages.

In Tokens We Trust: Exposing Superficial Editing and Unlearning with Adversarial Tokenization

Manit Baser, Aditya Nawal, Mohan Gurusamy

Under review

NDSS 2027

Security evaluation of post-release knowledge modification in open-weight language models.

PlaceCarry: Evaluating Placeholder-Based Semantic Carryover in Multi-Stage LLM Jailbreaks

Xiaodong Shi, Aditya Nawal, Mohan Gurusamy

Under review

ACL ARR, Aug 2026

Multi-turn safety evaluation of language models under indirect, context-dependent references.

Trust and Provenance in Multi-Agent Systems	In preparation
How agreement between collaborating agents affects the safety and provenance of what they produce.	
Adaptive Multi-Turn Adversarial Evaluation of Tool-Using Agents	In preparation
Extending retrieval-driven safety analysis to adaptive, multi-turn interaction over live web sources.	
Cost and Resource Integrity in LLM Agent Serving	In preparation
How untrusted context shapes the serving-time cost of agent workloads, and the policies that bound it.	

EXPERIENCE

National University of Singapore (NUS)	Jan 2025 – Present
<i>PhD Researcher, Department of Electrical and Computer Engineering</i>	
- Researching safety vulnerabilities in retrieval-augmented LLM agents and developing robust AI alignment methodologies. - Conducted mechanistic interpretability on LLM agents to inspect internal representations and belief shifts during adversarial persuasion.	
Microsoft	July 2022 – December 2024
<i>Software Development Engineer</i>	
- Engineered Azure OpenAI-assisted reasoning workflows to automate network failure diagnosis, enhancing consistency and resolution speed. - Developed production tooling using C# and React to integrate LLM capabilities into internal cloud infrastructure services. - Automated deployment workflows and onboarding pipelines for network operations utilizing infrastructure-as-code automation.	
Microsoft	Jan 2022 – June 2022
<i>Software Development Intern</i>	
- Designed and implemented a centralized service marketplace platform to streamline internal tool discovery and cloud service usage. - Consolidated fragmented software catalog services to improve developer experience and coordination across provider teams.	
Akamai Technologies	May 2021 – July 2021
<i>Software Development Intern</i>	
- Developed automated security tooling to test DNS query routing and validate packet handling integrity across security boundaries. - Configured simulated environments utilizing the TOR network and Onion routing protocols for anonymity assessment. - Engineered network packet captures (pcaps) replicating threat scenarios in the MITRE ATT&CK framework to test defense signatures.	

PROJECTS

Copilot Skill For Network Troubleshooting and Remediation	Azure OpenAI
- Developed an intelligent Copilot Skill using C# and Semantic Kernel for autonomous network fault diagnosis via natural language. - Built integration pipelines using Azure Functions to transform diagnostic telemetry into structured, actionable troubleshooting plans.	

TECHNICAL SKILLS

Core Languages & Frameworks: Python, C++ , PyTorch, JAX, TensorFlow, Git, LaTeX, C#, React, SQL
Research & Expertise: LLM Alignment (RLHF, DPO), AI Safety & Red Teaming, Adversarial Machine Learning, Mechanistic Interpretability, Multi-Agent Systems, RAG Security, Multilingual NLP
Agentic & LLM Frameworks: AutoGen, Semantic Kernel, Swarm, Magentic-One, LangChain, Hugging Face, OpenAI API

HONORS AND SERVICE

Leadership and Service: Led NSS BITS Pilani fundraising initiatives for underprivileged education and coordinated Microsoft IDC giving programs supporting child welfare, girl-child education, and disability-focused organizations.
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